
A
Directional signatures show discrimination in the expected cell-type direction
18,619 annotated immune cells, 4 pediatric high-grade gliomas (GSE227983); median AUROC 0.868
T Cell Cytotoxicity
Microglia Klemm2020
Microglia Core Homeostatic
NK Cell Activity
DAM KerenShaul2017
MgTAM Antunes2021
MHC Class II
MoTAM Antunes2021
Neutrophil Activation
MDM Klemm2020
M2 Macrophage
T Cell Exhaustion
Tregs Friebel2020
M1 Macrophage
Dendritic Cell Activation
Glioma Inflammatory Wang2017
0.075
0.0
0.2
0.4
0.6
0.8
1.0
AUROC for the cell type the signature is named for
myeloid signature
T / NK-cell signature
discordant — identifies T cells
B Cell class with highest mean expression
Top-expression genes (n)
0
20
40
60
80
100
Dendritic cell
Macrophage/
Microglia
T cell
NK cell
B/plasma
Endothelial
Mural
Immune highest (%)
98.0%
69.7%
Top 300 genes
Matched random
C Sample-1 (n=577)
Sample-2 (n=856)
Sample-3 (n=217)
Nearest immune-state centroid
High
Intermediate
Low
D Spot immune fraction by nearest-centroid class (Low, Intermediate, High)
Estimated immune fraction
0.0
0.2
0.4
0.6
0.8
Sample-1
Sample-2
Sample-3
E Spatial axis and mixed-cell profiles
Sample-1 (ρ=0.90; mixed 61%)
Sample-2 (ρ=0.76; mixed 59%)
Sample-3 (ρ=0.93; mixed 92%)
Estimated immune fraction
−1.50
−1.25
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
Spot immune-axis score